Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/20/2007
PI Name:	You, Li Ph.D.		
Project Title:	Theoretical studies of the generation, detection, and applications of massively entangled states of Bose condensed atoms		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30332-0430	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	A postdoc in the group, Dr. Peng Zhang was briefly supported on this program for about 6 months.		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	The experimental success in 1995 that produced the first dilute atomic Bose-Einstein condensate has led to an explosive growth of interests in this new type of degenerate quantum matter. As a testing ground for many fundamental issues in physics, in particular, in our understanding of interacting many body systems, atomic quantum gas has attracted much recent theoretical and experimental efforts. Ultimately, to grow beyond the novelty of being just another many body quantum mechanic phenomena, useful applications must be found that can take advantage of the macroscopic collective coherence of Bose condensed atoms, as well as the demonstrated abilities for the control of their interactions and the manipulation of their quantum states. At the frontier of atomic quantum gas research, a substantial topic of current interest is the creation of quantum correlated states of condensed atoms for several potential applications ranging from quantum computation to high precision quantum limited atom interferometry. In this proposal, we will investigate in detail the generation, detections		

Task Description:	and application of quantum correlated states of Bose condensed atoms. We will focus our efforts on two particular types of states that display inseparable quantum correlations: spin squeezed states and maximally entangled states. We will also study their relevance to several potential applications of trapped quantum gases. Our primary scientific objective to address the fundamental understanding of the proposed topics. We will develop new theoretical ideas and provide explicit estimate for parameters and conditions relevant to their technological applications. We expect our program will indirectly benefit experimental groups. To understand non-classical correlation and/or entanglement properties of Bose condensed atoms, and to develop effective means for their generation and detection. Spin squeezing and entanglement in an ensemble of fixed numbers of atoms can be used to enhance signal to noise ratios of certain atomic population based measurements, particularly for the atomic clock/frequency measurements based on the Ramsey interference scheme.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study constitutes an important part of the theoretical efforts in support of improved atomic clock technology.
Task Progress:	We have made two significant progresses during this reporting period. First, we have developed a protocol for the measurement of certain collective operators for a multi-party quantum state. More specifically, we implemented a party check operation, that is a key step for realizing the Heisenberg limited sensing capabilities of massively entangled states. It turns out that our protocol can be directly implemented with present ion trap technologies. A paper on this work has been published at Physical Review Letters. Second, we have made a significant step towards defining a quantum correlation measure among an arbitrary number of partitions of a Multi-qubit quantum state. This is a significant theoretical breakthrough as it allows for the first time a consistent comparison and calibration of quantum entanglement resources inside an arbitrary N-qubit quantum state. Previous measures can only be applied to measure these inseparable resources when the N-qubit system is partitioned into two parties only. A paper has now been submitted to Physical Review Letters. During this last phase of no-cost extension after the cancellation of our program, we have summarized and published a few articles. A postdoc in the group, Dr. Peng Zhang was briefly supported on this program for about 6 months.
Bibliography Type:	Description: (Last Updated: 08/26/2021)
Articles in Peer-reviewed Journals	Zeng B, Zhou DL, You L. "Measuring the parity of an N-qubit state." Phys Rev Lett. 2005 Sep 9;95(11):110502. PMID: 16196991 , Sep-2005
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